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Theory and Methodology

A comparison of heuristics and relaxations for the Capacitated Plant Location Problem

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Abstract: Approaches proposed in the literature for the Capacitated Plant Location Problem are compared. The comparison is based on new theoretical and computational results. The main emphasis is on relaxations. In particular, dominance relations among the various relaxations found in the literature are identified. In the computational study, the relaxations are compared as a function of various characteristics of the test problems. Several of these relaxations can be used to generate heuristic feasible solutions that are better than the classical greedy or interchange heuristics, both in computing time and in the quality of the solutions found.

Keywords: Location theory, capacitated facility location, Lagrangian relaxation, Lagrangian heuristic

1. Introduction

For many organizations, locational decisions are of utmost importance. The location of plants, such as warehouses or factories, affects the cost of supplying commodities to clients through transportation costs and the fixed costs of operating the plants. These costs vary with the location and size of the plants. The various questions associated with facility location have given rise to a vast literature, including special issues of *EJOR*. The reader is referred to the survey of Francis, McGinnis and White [12] for background material on locational decisions.

In this paper, we concentrate on one model known as the capacitated plant location problem. This problem alone has generated an important literature. In fact, so many different approaches have been proposed that it has become difficult for researchers and practitioners to know how they compare. We perform a systematic study of the relative quality of the bounds used in branch-and-bound algorithms in the literature. The reason for focusing on the bounds is that they are the overriding factor in the success of a branch-and-bound algorithm. When it comes to heuristics, researchers typically compare different variants of the greedy or interchange heuristics. We prefer 'Lagrangian heuristics'. We show here that they tend to provide much better solutions without being more computationally intensive.

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The Capacitated Plant Location Problem (CPLP) can be formulated as the mixed integer linear program

$$Z = \min \quad \sum_i \sum_j c_{ij} x_{ij} + \sum_j f_j y_j$$

subject to

$$\sum_j x_{ij} = 1 \quad \text{for all } i, \tag{D}$$

$$\sum_i d_i x_{ij} \leqslant s_j y_j \quad \text{for all } j, \tag{C}$$

$$0 \leqslant x_{ij} \leqslant 1, \quad 0 \leqslant y_j \leqslant 1 \quad \text{for all } i \text{ and } j, \tag{N}$$

$$y_j \text{ integer} \quad \text{for all } j. \tag{I}$$

In this formulation, i indexes the clients and j the sites where facilities can be located (both the set of clients and the set of sites are assumed to be finite); c_{ij} is the transportation cost incurred if all of client i's demand is met from facility j and f_j is the fixed cost of operating facility j; $d_i > 0$ is the demand of client i, and $s_j > 0$ is the capacity of facility j, if it is open. The decisions to be made are represented by the variables y_j and x_{ij}, namely $y_j = 1$ if facility j is open and 0 otherwise, and x_{ij} is the fraction of the demand of client i met from facility j. The constraints (D) are the demand constraints (for each client, all the demand must be met), (C) are the capacity constraints (no facility can supply more than its capacity), (N) are the nonnegativity and simple upper bound constraints and (I) the integrality constraints.

The applications of this model are not limited to plant location. For example, the same mathematical model is appropriate for making optimal lot sizing decisions in production planning [24,29]. The model CPLP has also been proposed in the context of telecommunication network design [23,26], machine replacement and as the basis for a heuristic in vehicle routing when vehicles have unequal capacities (see [31]).

The formulation of the CPLP can be enriched by simple valid inequalities. For example, the variable upper bound constraints

$$x_{ij} \leqslant y_j \quad \text{for all } i, j, \tag{B}$$

and the total demand constraint

$$\sum_j s_j y_j \geqslant \sum_i d_i, \tag{T}$$

are implied by the constraints (D), (C), (N), (I). To see that (B) is valid for CPLP, note that $y_j = 0$ or 1 by constraint (I). When $y_j = 1$, (B) follows from $x_{ij} \leqslant 1$ in constraint (N). When $y_j = 0$, (C) implies that $x_{ij} = 0$ (using $x_{ij} \geqslant 0$ and $d_i > 0$) and therefore (B) holds. To see that (T) is valid, sum up the constraints (C):

$$\sum_j \sum_i d_i x_{ij} \leqslant \sum_j s_j y_j.$$

Now, using (D), the constraint (T) follows.

Various lower bounds on Z are obtained by relaxing subsets of the constraints (D), (C), (N), (I), (B) and (T) either completely or in a Lagrangian fashion. We use the following notation. If a set of constraints, say (S), is relaxed completely the resulting bound is denoted by Z^{S}. If constraints other than (I) are relaxed in a Lagrangian fashion, the corresponding Lagrangian dual bound is denoted by Z_{S}. See, for example, [28] for an introduction to Lagrangian duality. For example, with the above notation,

$$Z_{\mathrm{D}} = \max_u \min_{x,\, y} \quad \sum_i \sum_j c_{ij} x_{ij} + \sum_j f_j y_j - \sum_i u_i \left(\sum_j x_{ij} - 1 \right)$$

subject to (C), (N), (I), (B), (T).

These lower bounds provide the basis for numerous algorithms. For example, the following relaxations have been used in branch-and-bound algorithms.

- Z^{BI} This relaxation is the so-called weak linear programming relaxation of CPLP. It was used by Sa [30] and improved by Akinc and Khumawala [1] using ad hoc rules.
- Z^{I} This is the strong linear programming relaxation of the problem. It was used by Davis and Ray [7], Guignard and Spielberg [20], and Baker [2].
- $Z^{\mathrm{T}}_{\mathrm{DC}}$ Geoffrion and McBride [15].
- $Z^{\mathrm{B}}_{\mathrm{D}}$ Nauss [27].
- Z'_{D} Christofides and Beasley [5]. In this relaxation, the total demand constraint (T) is replaced by $\sum_j y_j \geqslant K$ where K is the smallest integer such that the sum of the K largest s_j's is at least $\sum_i d_i$.
- $Z^{\mathrm{T}}_{\mathrm{C}}$ Van Roy [33].
- Z_{C} Van Roy [34], Guignard and Kim [17].

Let $(S_1),\ldots,(S_k)$ represent sets of equality or inequality constraints. We use the notation $P(S_1\ldots S_k)$ to denote the region defined by the constraints $(S_1),\ldots,(S_k)$.

Remark 1.

$$P(S_1\ldots S_k)=P(S_1)\cap\cdots\cap P(S_k).$$

We have to treat the integrality requirement (I) differently. We denote by $\mathrm{Conv}(S_1\ldots S_k I)$ the convex hull of the feasible solutions to the constraints $(S_1),\ldots,(S_k)$, (I). In this paper, such sets will always be closed.

Remark 2.

$$\mathrm{Conv}(S_1S_2\mathrm{I})\subseteq P(S_1)\cap\mathrm{Conv}(S_2\mathrm{I})\subseteq P(S_1S_2).$$

With this notation we can write

$$Z=\min cx+fy,$$
$$(x,\ y)\in\mathrm{Conv}(\mathrm{DCNIBT}).$$

Using an important theorem of Geoffrion [14], we can also express the Lagrangian dual bounds with this notation. For example,

$$Z_{\mathrm{D}}=\min cx+fy,$$
$$(x,\ y)\in P(\mathrm{D})\cap\mathrm{Conv}(\mathrm{CNIBT}).$$

In general, the rule to obtain $Z_{\mathrm{R}}^{\mathrm{S}}$ is to discard (S) altogether and to move (R) out of the group of convexified constraints.

Recently, a new type of relaxation has been introduced by several authors. See [18] for a survey and new results. The method, known as *variable splitting,* can be defined in general for (mixed) integer programs. It is obtained by rewriting the program

$$\begin{aligned} Z=\min\ & cu,\\ & A_1u\leqslant b_1, && (\mathrm{S}_1)\\ & A_2u\leqslant b_2, && (\mathrm{S}_2)\\ & u\in X, && (\Sigma)\end{aligned}$$

where (Σ) contains the integrality constraints (I), as

$$\begin{aligned} Z=\min\ & cu,\\ & A_1u\leqslant b_1,\\ & u\in X,\\ & A_2v\leqslant b_2,\\ & v\in X,\quad u=v,\end{aligned}$$

and by performing a Lagrangian relaxation of the equalities $u=v$. The Lagrangian dual bound obtained this way has the value

$$Z_{S_1/S_2}=\min cu,$$
$$u\in\mathrm{Conv}(S_1\Sigma)\cap\mathrm{Conv}(S_2\Sigma).$$

See [18] for a proof of this result. As a consequence, $Z_{S_1}\leqslant Z_{S_1/S_2}$, i.e., the bound obtained by variable splitting is always at least as tight as the corresponding Lagrangian bound.

Our motivation for this study is twofold, theoretical and computational. On the theoretical side, we rank relaxations according to the tightness of the bounds that they yield. This analysis is performed in Section 2. Several inequalities given in that section are classical, but we have had a few surprises. For example, consider Z_{C}, Z_{TC} and $Z_{\mathrm{C}}^{\mathrm{T}}$. It is easy to show that $Z_{\mathrm{C}}\geqslant Z_{\mathrm{TC}}\geqslant Z_{\mathrm{C}}^{\mathrm{T}}$ and Van Roy [34] and Guignard and Kim [17] have noticed that the bound Z_{C} is often strictly stronger than $Z_{\mathrm{C}}^{\mathrm{T}}$ in practice. However Z_{C} is also harder to compute than $Z_{\mathrm{C}}^{\mathrm{T}}$ and, in both references, the Lagrangian relaxation of (T) is introduced. Indeed, it seems that computing Z_{TC} is a good compromise between bound quality and ease of solution. We show that, in fact, $Z_{\mathrm{TC}}=Z_{\mathrm{C}}^{\mathrm{T}}$. In other words, the quality of Z_{C} is lost in the same proportion whether (T) is relaxed completely or in a Lagrangian fashion. Note that the solution of Z_{TC} by subgradient optimization may still have merits as it is possible that it converges faster than a subgradient approach applied directly to $Z_{\mathrm{C}}^{\mathrm{T}}$. We also prove the important inequality $Z_{\mathrm{D}}\leqslant Z_{\mathrm{C}}$ which does not seem to be well-known, although it is mentioned in [34].

Another surprising result from Section 2 is that variable splitting does not yield stronger bounds than the best Lagrangian bounds, for the CPLP. For example, in a recent paper, Barcelo, Fernandez, and Jörnsten [3] propose the split $Z_{\mathrm{D/CBT}}$. We show in Section 2 that $Z_{\mathrm{D/CBT}}=Z_{\mathrm{D}}$. In [19], another split is proposed, namely $Z_{\mathrm{CT/DB}}$. They show $Z_{\mathrm{CT/DB}}\geqslant Z_{\mathrm{C}}^{\mathrm{T}}$ and $Z_{\mathrm{CT/DB}}\geqslant Z_{\mathrm{D}}$. In this paper, we show that $Z_{\mathrm{CT/DB}}\leqslant Z_{\mathrm{C}}$.

Section 3 analyzes the complexity of computing the different bounds found in Section 2. We also consider the complexity of the problem CPLP itself under two different assumptions on the capacities of the facilities.

Section 4 contains our computational experi-

ence with various relaxations. Nearly all the researchers in the field have tested their algorithms on the same set of data, the famous Kuehn and Hamburger test problems as adapted by Khumawala [22]. These test problems have been a great help in comparing the various solution methods. At the same time, it should be noted that this test set is relatively small and has a very special property: all the facilities have the same capacity and operating cost. Therefore it seems appropriate to conduct a computational study on a different and larger set of test problems. The problems we consider have varying capacities and operating costs but, for each facility, the operating cost is correlated to the capacity according to an increasing concave function (economies of scale). The duality gap yielded by various relaxations appears to be sensitive to characteristics of the data such as total capacity versus total demand. To demonstrate such relationships we perform our computational experiments as a function of $\rho = \sum_j s_j / \sum_i d_i$. Keeping everything else equal, we scale the s_j's in order to get different values of ρ. For example we find that adding the total demand constraint (T) improves the relaxations significantly when ρ is small.

Our computational experience also covers some existing heuristics. Here as well we have had surprises. A number of heuristics which perform well on Kuehn and Hamburger problems have a poor performance on our test problems. We recommend the use of Lagrangian heuristics rather than the usual greedy or interchange-type heuristics. Consider any iterative method (such as subgradient optimization) to compute Z_D. For each u^t, $t \geqslant 1$, generated during the iterative process, the subproblem in x and y

$$\min_{x,\,y} \quad \sum_i \sum_j c_{ij} x_{ij} + \sum_j f_j y_j - \sum_i u_i^t \left(\sum_j x_{ij} - 1 \right)$$

$$\text{subject to} \quad \text{(C), (N), (I), (B), (T),}$$

is solved. The vector (x_D, y_D) so obtained produces a Lagrangian bound, but the vector y_D can also be used to generate a heuristic solution. Indeed, given y_D, one can solve a transportation problem to get a feasible solution $\bar{x}_D$. Thus at each iteration of the iterative process, a heuristic solution $(\bar{x}_D, y_D)$ can be generated for the CPLP. The best solution found in 50 or so subgradient iterations can be used as a heuristic. We find that this heuristic is just as fast and gives much better solutions than the more commonly used greedy-interchange approach. Instead of Z_D, one can also use Z_C^T or Z_C as the basis for the heuristic but we do not recommend it, as little additional accuracy is gained at great computational cost.

2. Relative strength of the relaxations

In this section we present two theorems. In Theorem 1, we compare the strength of the various linear programming relaxations and Lagrangian duals of CPLP. We compare all the possible bounds that can arise in this way, except that we do not relax (N) (there is no computational gain in doing so), nor do we consider complete relaxations of (C) or (D) since we feel that such bounds would be of little interest in solving CPLP (Z^C and Z^D are NP-hard to compute, yet they lack important aspects of CPLP.) Specifically, we consider all the combinations where (T) and (B) are present, relaxed completely or in a Lagrangian fashion, and where (C) and (D) are present or relaxed in a Lagrangian fashion, i.e. $3 \times 3 \times 2 \times 2 = 36$ relaxations, as well as the LP relaxations Z^I, Z^{BI}, Z^{TI} and Z^{TBI}. Other combinations where (I) is relaxed completely are omitted as it is well-known that they yield the same LP bounds. In addition, we consider the relaxation Z'_D of Christofides and Beasley [5] defined in Section 1 above. Theorem 1 states that these 41 relaxations only yield 7 genuinely different bounds. It is interesting to note that 6 of these 7 bounds have already been proposed in the literature as the basis for branch-and-bound algorithms.

In Theorem 2, we discuss relaxations that can be obtained by variable splitting. We show that, for the CPLP, the bounds arising from these relaxations are at least as strong as Z_D but no stronger than Z_C.

Theorem 1. *The following inequalities hold*:

$$\begin{aligned} &Z^{BI} \leqslant Z^I \leqslant Z_C^T \leqslant Z_C \leqslant Z, \\ &Z^I \leqslant Z'_D \leqslant Z_D \leqslant Z_C, \\ &Z^{BI} \leqslant Z_C^B \leqslant Z_D. \end{aligned} \tag{2.1}$$

Every inequality in (2.1) *is strict for at least one instance of CPLP, and no other inequality exists among the values of* (2.1) *except those derived by transitivity.*

The following equalities hold:

$$Z = Z_{\mathrm{B}} = Z^{\mathrm{B}} = Z_{\mathrm{T}} = Z^{\mathrm{T}} = Z_{\mathrm{TB}} = Z_{\mathrm{T}}^{\mathrm{B}} = Z_{\mathrm{B}}^{\mathrm{T}} = Z^{\mathrm{TB}}, \tag{2.2}$$

$$Z_{\mathrm{D}} = Z_{\mathrm{DC}} = Z_{\mathrm{BD}} = Z_{\mathrm{BC}} = Z_{\mathrm{BDC}} = Z_{\mathrm{D}}^{\mathrm{B}}, \tag{2.3}$$

$$Z_{\mathrm{C}}^{\mathrm{B}} = Z_{\mathrm{DC}}^{\mathrm{B}}, \tag{2.4}$$

$$Z_{\mathrm{C}}^{\mathrm{T}} = Z_{\mathrm{TC}}, \tag{2.5}$$

$$Z^{\mathrm{I}} = Z^{\mathrm{TI}} = Z_{\mathrm{TD}} = Z_{\mathrm{D}}^{\mathrm{T}} = Z_{\mathrm{TDC}} = Z_{\mathrm{DC}}^{\mathrm{T}} = Z_{\mathrm{BTC}} = Z_{\mathrm{BC}}^{\mathrm{T}} = Z_{\mathrm{BTDC}} = Z_{\mathrm{BDC}}^{\mathrm{T}} = Z_{\mathrm{BTD}} = Z_{\mathrm{BD}}^{\mathrm{T}} = Z_{\mathrm{TD}}^{\mathrm{B}} = Z_{\mathrm{D}}^{\mathrm{BT}}, \tag{2.6}$$

$$Z^{\mathrm{BI}} = Z^{\mathrm{TBI}} = Z_{\mathrm{TC}}^{\mathrm{B}} = Z_{\mathrm{C}}^{\mathrm{TB}} = Z_{\mathrm{TDC}}^{\mathrm{B}} = Z_{\mathrm{DC}}^{\mathrm{TB}}. \tag{2.7}$$

Proof. In (2.1), the inequalities $Z^{\mathrm{BI}} \leqslant Z^{\mathrm{I}}$ and $Z_{\mathrm{C}}^{\mathrm{T}} \leqslant Z_{\mathrm{C}} \leqslant Z$ are immediate by relaxation. $Z_{\mathrm{D}}' \leqslant Z_{\mathrm{D}}$ follows from the fact that the constraint used in Z_{D}' in place of (T) is implied by (I) and (T). Using the property that Lagrangian duals are at least as strong as the corresponding linear programming relaxation [14], we get $Z^{\mathrm{BI}} \leqslant Z_{\mathrm{C}}^{\mathrm{B}}$, $Z^{\mathrm{IT}} \leqslant Z_{\mathrm{D}}'$ and $Z^{\mathrm{IT}} \leqslant Z_{\mathrm{C}}^{\mathrm{T}}$. This implies $Z^{\mathrm{I}} \leqslant Z_{\mathrm{D}}'$ and $Z^{\mathrm{I}} \leqslant Z_{\mathrm{C}}^{\mathrm{T}}$ provided we can show $Z^{\mathrm{I}} = Z^{\mathrm{IT}}$. This will be done in (2.6). $Z_{\mathrm{C}}^{\mathrm{B}} \leqslant Z_{\mathrm{D}}$ follows from $Z_{\mathrm{DC}}^{\mathrm{B}} \leqslant Z_{\mathrm{D}}$ provided we can show $Z_{\mathrm{C}}^{\mathrm{B}} = Z_{\mathrm{DC}}^{\mathrm{B}}$. This will be done in (2.4). Finally $Z_{\mathrm{D}} \leqslant Z_{\mathrm{C}}$ follows from $Z_{\mathrm{DC}} \leqslant Z_{\mathrm{C}}$ provided we can show $Z_{\mathrm{D}} = Z_{\mathrm{DC}}$. This will be done in (2.3). So, to show the validity of the theorem, it remains to prove the equalities (2.2)–(2.7) and to exhibit instances showing that all the inequalities in (2.1) can be strict and that no other inequality exists. These results are provided in the next two lemmas. □

Lemma 1. *The equalities* (2.2)–(2.7) *hold.*

Proof. We give polyhedral proofs of these results. See [31] for algebraic proofs.

The constraints (T) and (B) can be derived from (D), (C), (I) and (N) as shown in Section 1. Therefore

$$\mathrm{Conv}(\mathrm{DCIN}) = \mathrm{Conv}(\mathrm{TBDCIN}),$$

proving (2.2).

To prove (2.3), we show that $Z_{\mathrm{D}} = Z_{\mathrm{BDC}}$ and then that $Z_{\mathrm{D}} = Z_{\mathrm{D}}^{\mathrm{B}}$.

$$Z_{\mathrm{D}} = \min cx + fy, \quad (x, y) \in P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{BCINT}),$$

and

$$Z_{\mathrm{BDC}} = \min cx + fy, \quad (x, y) \in P(\mathrm{BDC}) \cap \mathrm{Conv}(\mathrm{INT}).$$

So, the equality $Z_{\mathrm{D}} = Z_{\mathrm{BDC}}$ holds if we can prove the polyhedral result

$$P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{BCINT}) = P(\mathrm{BDC}) \cap \mathrm{Conv}(\mathrm{INT}). \tag{2.8}$$

Since $P(\mathrm{BDC}) = P(\mathrm{D}) \cap P(\mathrm{BC})$ by Remark 1, we only have to show the following.

Claim.

$$\mathrm{Conv}(\mathrm{BCINT}) = P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT}). \tag{2.9}$$

Proof. By Remark 2,

$$\mathrm{Conv}(\mathrm{BCINT}) \subseteq P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT}).$$

Now we prove

$$P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT}) \subseteq \mathrm{Conv}(\mathrm{BCINT}).$$

Consider any extreme point of $P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT})$, say $(\bar{x}, \bar{y})$. If $\bar{y}$ has only 0, 1 coordinates, then $(\bar{x}, \bar{y}) \in \mathrm{Conv}(\mathrm{BCINT})$ and the claim is proved. So assume that $\bar{y}$ has at least one fractional component. We will show that this cannot occur. Denote by $\mathrm{Conv}_y(\mathrm{INT})$ the projection of the polyhedron Conv(INT) onto the y-space. Since $\mathrm{Conv}_y(\mathrm{INT})$ has only integral vertices, $\bar{y}$ is not a vertex and therefore $\bar{y} = \frac{1}{2}y^1 + \frac{1}{2}y^2$ where $y^1, y^2 \in \mathrm{Conv}_y(\mathrm{INT})$ and $y^1 \neq y^2$. If $\bar{y}_j = 0$, set $x_{ij}^1 = x_{ij}^2 = 0$. Else, set

$$x_{ij}^k = \bar{x}_{ij}\left(y_j^k/\bar{y}_j\right) \quad \text{for } k = 1, 2.$$

We have $(x^k, y^k) \in P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT})$ for $k = 1, 2$ and

$$(\bar{x}, \bar{y}) = \tfrac{1}{2}(x^1, y^1) + \tfrac{1}{2}(x^2, y^2).$$

This contradicts the assumption that $(\bar{x}, \bar{y})$ is an extreme point of $P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{INT})$, completing the proof of the claim. □

Now we prove $Z_{\mathrm{D}} = Z_{\mathrm{D}}^{\mathrm{B}}$ by showing

$$P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{BCINT}) = P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{CINT}). \tag{2.10}$$

This relation follows from the fact that (C), (I) and (N) imply (B). This completes the proof of (2.3).

To prove (2.4), we show that

$$P(\mathrm{C}) \cap \mathrm{Conv}(\mathrm{DINT}) = P(\mathrm{DC}) \cap \mathrm{Conv}(\mathrm{INT}). \tag{2.11}$$

Since $P(\mathrm{DC}) = P(\mathrm{C}) \cap P(\mathrm{D})$ by Remark 1, it suffices to show

$$P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{INT}) = \mathrm{Conv}(\mathrm{DINT}).$$

The constraints (D), (I), (N) and (T) completely separate into the constraints involving x and those involving y. The convexification step only involves the variables y and, therefore, the result follows. This completes the proof of (2.4).

The equality (2.5) follows from

$$P(\mathrm{C}) \cap \mathrm{Conv}(\mathrm{DINB}) = P(\mathrm{TC}) \cap \mathrm{Conv}(\mathrm{DINB}).$$

To see this, note that (T) is a consequence of (C) and (D) and therefore

$$\begin{aligned} &P(\mathrm{C}) \cap \mathrm{Conv}(\mathrm{DINB}) \\ &\quad = P(\mathrm{T}) \cap P(\mathrm{C}) \cap \mathrm{Conv}(\mathrm{DINB}). \end{aligned}$$

Now we show (2.6). First, $Z^{\mathrm{I}} = Z^{\mathrm{TI}}$ since (T) follows from (C) and (D). In (2.6), the only two Lagrangian duals that might not be at least as strong as $Z^{\mathrm{I}} = Z^{\mathrm{TI}}$ are $Z^{\mathrm{B}}_{\mathrm{TD}}$ and $Z^{\mathrm{BT}}_{\mathrm{D}}$. Next, we give a polyhedral proof of $Z^{\mathrm{BT}}_{\mathrm{D}} = Z^{\mathrm{B}}_{\mathrm{TD}} = Z_{\mathrm{TD}}$. We have

$$\begin{aligned} P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{CIN}) &= P(\mathrm{TD}) \cap \mathrm{Conv}(\mathrm{CIN}) \\ &= P(\mathrm{TD}) \cap \mathrm{Conv}(\mathrm{CINB}), \end{aligned}$$

where the first equality holds because (D) and (C) imply (T) and the second equality holds because (C), (I), (N) imply (B). To complete the proof of (2.6) it suffices to show $Z_{\mathrm{TD}} = Z^{\mathrm{I}}$ and $Z_{\mathrm{BTC}} = Z^{\mathrm{I}}$, since all the other Lagrangian duals in (2.6) are dominated by Z_{TD} or Z_{BTC}.

To show $Z_{\mathrm{TD}} = Z^{\mathrm{I}}$, we prove the polyhedral result

$$P(\mathrm{TD}) \cap \mathrm{Conv}(\mathrm{CINB}) = P(\mathrm{TDCNB}). \tag{2.12}$$

Using $P(\mathrm{TDCNB}) = P(\mathrm{TD}) \cap P(\mathrm{CNB})$, it remains to show $\mathrm{Conv}(\mathrm{CINB}) = P(\mathrm{CNB})$. The argument made to prove (2.9) holds as well when (T) is not present. So

$$\mathrm{Conv}(\mathrm{BCIN}) = P(\mathrm{BC}) \cap \mathrm{Conv}(\mathrm{IN}).$$

The extreme points of the hypercube (N) have 0, 1 components. So $\mathrm{Conv}(\mathrm{IN}) = P(\mathrm{N})$. This implies $\mathrm{Conv}(\mathrm{BCIN}) = P(\mathrm{BCN})$ and (2.12) is proved. To show $Z_{\mathrm{BTC}} = Z^{\mathrm{I}}$, we prove

$$P(\mathrm{BTC}) \cap \mathrm{Conv}(\mathrm{DIN}) = P(\mathrm{BTCDN}). \tag{2.13}$$

This follows from

$$\begin{aligned} \mathrm{Conv}(\mathrm{DIN}) &= P(\mathrm{D}) \cap \mathrm{Conv}(\mathrm{IN}) \\ &= P(\mathrm{D}) \cap P(\mathrm{N}) = P(\mathrm{DN}), \end{aligned}$$

where the first equality holds because the variables x in the constraints (D), and (N) separate completely from the integer variables y.

Finally consider (2.7). $Z^{\mathrm{BI}} = Z^{\mathrm{TBI}}$ holds as (T) follows from (C) and (D). So it suffices to show $Z^{\mathrm{BI}} = Z^{\mathrm{B}}_{\mathrm{TC}}$. To prove it, we need to show

$$P(\mathrm{TCDN}) = P(\mathrm{TC}) \cap \mathrm{Conv}(\mathrm{DIN}).$$

This follows from $\mathrm{Conv}(\mathrm{DIN}) = P(\mathrm{DN})$, which we just proved above. □

To complete the proof of the theorem, it suffices to exhibit instances showing that all the inequalities can be strict.

Lemma 2. *There exist instances of the CPLP showing that each of the following inequalities can hold:*

$$Z^{\mathrm{BI}} < Z^{\mathrm{I}}, \tag{2.14}$$

$$Z'_{\mathrm{D}} < Z^{\mathrm{B}}_{\mathrm{C}}, \tag{2.15}$$

$$Z^{\mathrm{T}}_{\mathrm{C}} < Z^{\mathrm{B}}_{\mathrm{C}}, \tag{2.16}$$

$$Z^{\mathrm{T}}_{\mathrm{C}} < Z'_{\mathrm{D}}, \tag{2.17}$$

$$Z_{\mathrm{D}} < Z^{\mathrm{T}}_{\mathrm{C}}, \tag{2.18}$$

$$Z^{\mathrm{B}}_{\mathrm{C}} < Z^{\mathrm{I}}, \tag{2.19}$$

$$Z_{\mathrm{C}} < Z. \tag{2.20}$$

The reader can verify that these conditions imply that every inequality in (2.1) can be strict and that no other inequality exists among the bounds of (2.1), except those derived by transitive closure.

Proof. First we give an example that satisfies (2.14) and (2.15).

$$(c_{ij}) = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}, \quad f = (2 \quad 3),$$

$$s = (3 \quad 5),$$

$$d = (2 \quad 3).$$

To compute Z^{BI}, we solve the corresponding linear program. Its optimum solution is $y_1 = 1$, $y_2 = \frac{2}{5}$, $x_{11} = 1$, $x_{12} = 0$, $x_{21} = \frac{1}{3}$ and $x_{22} = \frac{2}{3}$, yielding the bound $Z^{\mathrm{BI}} = \frac{16}{5}$. Now consider Z^{I}. Its optimum solution is the same as for Z^{BI}, except that $y_2 = \frac{2}{3}$. This yields $Z^{\mathrm{I}} = 4$. Thus this example shows that (2.14) can occur.

Note that, for this instance, the constraint (T′) used in Z'_{D} is $y_1 + y_2 \geqslant 1$ and therefore does not strengthen the relaxation $Z^{\mathrm{T}}_{\mathrm{D}}$. Since we have proved that $Z^{\mathrm{T}}_{\mathrm{D}} = Z^{\mathrm{I}}$, we get $Z'_{\mathrm{D}} = Z^{\mathrm{T}}_{\mathrm{D}} = Z^{\mathrm{I}} = 4$.

Now we compute $Z^{\mathrm{B}}_{\mathrm{C}}$. Consider the Lagrangian relaxation

$$Z^{\mathrm{B}}_{\mathrm{C}v} = \min \sum_i \sum_j c_{ij}x_{ij} + \sum_j f_j y_j + \sum_j v_j \Big(\sum_i d_i x_{ij} - s_j y_j \Big)$$

subject to

$$\sum_j x_{ij} = 1 \quad \text{for all } i,$$

$$\sum_j s_j y_j \geqslant \sum_i d_i,$$

$$x_{ij} \geqslant 0 \quad \text{for all } i,\ j,$$

$$y_j \in \{0, 1\} \quad \text{for all } j.$$

If we set $v_1 = \frac{2}{3}$, $v_2 = 0$, an optimum solution of the Lagrangian relaxation is $y_1 = y_2 = 1$, $x_{11} = 1$, $x_{12} = 0$, $x_{21} = \frac{1}{3}$, $x_{22} = \frac{2}{3}$. The corresponding value is $Z^{\mathrm{B}}_{\mathrm{C}v} = 5$. Since we also have $Z = 5$, we deduce $Z^{\mathrm{B}}_{\mathrm{C}} = Z = 5$. Therefore the strict inequality (2.15) is satisfied by this instance of the CPLP.

To prove that (2.16) and (2.17) can hold, consider the previous example, where $s_2 = 5$ is changed into $s_2 = 4$. It is easy to check that this modification does not affect $Z^{\mathrm{B}}_{\mathrm{C}}$ and Z^{I}. But, now, the constraint (T′) used in Z'_{D} is $y_1 + y_2 \geqslant 2$ and therefore $Z'_{\mathrm{D}} = Z^{\mathrm{B}}_{\mathrm{C}} = 5$. Next we compute $Z^{\mathrm{T}}_{\mathrm{C}} = \max_{v \geqslant 0} Z^{\mathrm{T}}_{\mathrm{C}v}$, where

$$Z^{\mathrm{T}}_{\mathrm{C}v} = \min \sum_i \sum_j c_{ij}x_{ij} + \sum_j f_j y_j + \sum_j v_j \Big(\sum_i d_i x_{ij} - s_j y_j \Big)$$

subject to

$$\sum_j x_{ij} = 1 \quad \text{for all } i,$$

$$0 \leqslant x_{ij} \leqslant y_j \quad \text{for all } i,\ j,$$

$$y_j \in \{0, 1\} \quad \text{for all } j.$$

This constraint set is totally unimodular as it is a simple plant location problem with two clients and two locations (see [4] for example). Therefore $Z^{\mathrm{T}}_{\mathrm{C}} = Z^{\mathrm{I}} = 4$. This proves that (2.16) and (2.17) can occur.

Next we give an example that satisfies (2.18).

$$(c_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad f = (1 \quad 1 \quad 1),$$

$$s = (3 \quad 3 \quad 3),$$

$$d = (1 \quad 1 \quad 1).$$

The optimum value of the linear programming relaxation $Z^{\mathrm{I}} = \frac{3}{2}$ is obtained by taking the solution $y_j = \frac{1}{2}$, $x_{ij} = \frac{1}{2}$ if $i \neq j$ and $x_{ij} = 0$ if $i = j$.

Note that, for this instance, $Z^{\mathrm{T}}_{\mathrm{D}} = Z_{\mathrm{D}}$ since each facility has enough capacity to meet the demand of all the clients and therefore the capacity constraints are automatically satisfied. Since we have proved in (2.6) that $Z^{\mathrm{T}}_{\mathrm{D}} = Z^{\mathrm{I}}$, we obtain $Z_{\mathrm{D}} = Z^{\mathrm{T}}_{\mathrm{D}} = Z^{\mathrm{I}} = \frac{3}{2}$.

Similarly $Z^{\mathrm{CT}} = Z$ since the capacity constraints are automatically satisfied. This shows that $Z^{\mathrm{CT}} = Z^{\mathrm{T}}_{\mathrm{C}} = Z = 2$.

Therefore this instance shows that possibility (2.18) can occur.

Finally we give an instance that satisfies (2.19) and (2.20).

$$(c_{ij}) = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{pmatrix}, \quad f = \left(\tfrac{3}{2} \quad \tfrac{3}{2} \quad \tfrac{3}{2}\right),$$

$$s = (3 \quad 3 \quad 3),$$

$$d = (2 \quad 2 \quad 2).$$

To prove (2.19), note that $Z^{\mathrm{B}}_{\mathrm{DC}} = Z^{\mathrm{B}}_{\mathrm{C}} \leqslant 3$ since the point $y_1 = y_2 = y_3 = \frac{2}{3}$ and $x_{ij} = 1$ if $i = j$ and 0 if $i \neq j$ belongs to $P(\mathrm{DC}) \cap \mathrm{Conv(INT)}$. On the other hand, $Z^{\mathrm{I}} = 4$ since

$$y_1 = y_2 = y_3 = \tfrac{2}{3}$$

and

$$x_{11} = x_{22} = x_{33} = \tfrac{2}{3}, \quad x_{13} = x_{21} = x_{32} = \tfrac{1}{3},$$

$$x_{12} = x_{23} = x_{31} = 0$$

is an optimum solution of the linear programming relaxation. To prove (2.20), note that $Z = \frac{9}{2}$ is obtained by setting $y_1 = y_2 = y_3 = 1$, $x_{ij} = 1$ if $i = j$

and 0 if $i \neq j$. For the solution of Z_C, consider $v_1 = v_2 = v_3 = 0$ in the Lagrangian relaxation Z_{Cv}. There are three alternate optimum solutions obtained from $y^1 = (0\ 1\ 1)$, $y^2 = (1\ 0\ 1)$ and $y^3 = (1\ 1\ 0)$ respectively. The corresponding subgradients are $(0\ -1\ 1)$, $(1\ 0\ -1)$ and $(-1\ 1\ 0)$. Their sum is the zero vector. This shows that $Z_C = 4$, proving (2.20).

This completes the proof of Lemma 2. □

We close this section with results about variable splitting. It will be useful to generalize the notation introduced in Section 1. If the set of inequalities (S_1) consists of the constraint sets $(S_1^1), \ldots, (S_1^k)$ and (S_2) consists of $(S_2^1), \ldots, (S_2^l)$, then the bound defined by the split (S_1), (S_2) and (Σ) is denoted by $Z_{S_1^1 \cdots S_1^k / S_2^1 \cdots S_2^l}$. As mentioned in the introduction, Guignard and Kim [18] showed that this value is obtained by minimizing $cx + fy$ over

$$(x, y) \in \mathrm{Conv}(S_1^1 \cdots S_1^k \Sigma) \cap \mathrm{Conv}(S_2^1 \cdots S_2^l \Sigma). \tag{2.21}$$

Let us consider all the possible splits of the sets of constraints (D), (C), (B), (T), (I) and (N) for CPLP. We make the following observations. First, note that (I) belongs to (Σ) by definition. We also assume that (N) belongs to (Σ) as no benefit can be gained by doing otherwise. There are 25 possible splits of (D), (C), (B), (T), (I) and (N) with the property that (I) and (N) belong to (Σ). If (S_1) or (S_2) is reduced to one of the sets (B), (T) or (B) ∪ (T), then we do not get a true relaxation of CPLP, as one of the subproblems is CPLP itself. Therefore the true relaxations have the property that (D) belongs to (S_1) and (C) belongs to (S_2) or vice versa. There are 9 such relaxations of CPLP. The next theorem shows that the bounds they produce are equal to Z_C, Z_D or to some intermediate value.

Theorem 2. *The following relations hold*:

$$Z_{T/*} = Z_{B/*} = Z_{BT/*} = Z, \tag{2.22}$$

where $*$ *stands for any subset of the remaining constraints*,

$$Z_{C/D} = Z_{C/DB} = Z_{C/DT} = Z_{C/DBT} = Z_C, \tag{2.23}$$

$$\max(Z_C^T, Z_D) \leqslant Z_{D/TC} = Z_{BD/TC} \leqslant Z_C, \tag{2.24}$$

and there are instances of CPLP showing that each of the inequalities $Z_C^T \leqslant Z_{D/TC}$, $Z_D \leqslant Z_{D/TC}$ *and* $Z_{D/TC} \leqslant Z_C$ *can be strict*,

$$Z_{D/BC} = Z_{D/TBC} = Z_{TD/BC} = Z_D. \tag{2.25}$$

Proof. (2.22) is immediate as one of the two subproblems is CPLP itself.

Next, we prove (2.23). Note that $Z_{C/D}$, $Z_{C/BD}$, $Z_{C/DT}$ and $Z_{C/DBT}$ are all at least as strong as Z_C, and $Z_{C/D}$ is the strongest of the four. To prove $Z_C = Z_{C/D}$ we show the polyhedral result

$$P(C) \cap \mathrm{Conv}(\mathrm{DINTB}) = \mathrm{Conv}(\mathrm{CINTB}) \cap \mathrm{Conv}(\mathrm{DINTB}).$$

This follows from

$$\mathrm{Conv}(\mathrm{CINTB}) = P(C) \cap \mathrm{Conv}(\mathrm{INTB})$$

as a consequence of (2.9), and from

$$\mathrm{Conv}(\mathrm{DINTB}) \subseteq \mathrm{Conv}(\mathrm{INTB}).$$

To prove (2.24), we show first that $Z_{D/TC} = Z_{BD/TC} \leqslant Z_C$. We have $Z_{BD/TC} \leqslant Z_{D/TC} \leqslant Z_{D/C} = Z_C$, where the two inequalities hold by relaxation and the equality has just been proved above. Next, we show $Z_{BD/TC} = Z_{D/TC}$. This holds since

$$\mathrm{Conv}(\mathrm{CINT}) \cap \mathrm{Conv}(\mathrm{DINB}) = \mathrm{Conv}(\mathrm{CINTB}) \cap \mathrm{Conv}(\mathrm{DINB})$$

follows from the fact that (C), (I) and (N) imply (B). The inequalities $Z_D \leqslant Z_{D/TC}$ and $Z_{TC} \leqslant Z_{D/TC}$ are a consequence of the fact that variable splitting produces a bound at least as strong as the corresponding Lagrangian bound.

To complete the proof of (2.24), we need to exhibit instances of CPLP where the inequalities are strict. It was shown in the proof of Lemma 2 that $Z_C^T = 4$ and $Z = Z_D = 5$ for the following instance.

$$(c_{ij}) = \begin{pmatrix} 0 & 3 \\ 0 & 0 \end{pmatrix}, \quad f = (2\ \ 3),$$

$$s = (3\ \ 4),$$

$$d = (2\ \ 3).$$

The relation $Z_D \leqslant Z_{D/TC}$ implies $Z_{D/TC} = 5$.

Therefore the inequality $Z_C^T \leqslant Z_{D/TC}$ can be strict. For the instance

$$(c_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad f = (1 \quad 1 \quad 1),$$

$$s = (3 \quad 3 \quad 3),$$

$$d = (1 \quad 1 \quad 1),$$

it was shown in the proof of Lemma 2 that $Z_D = \frac{3}{2}$ and $Z = Z_{TC} = 2$. Since $Z_{TC} \leqslant Z_{D/TC}$, we have $Z_{D/TC} = 2$, showing that $Z_D \leqslant Z_{D/TC}$ can be a strict inequality. Finally, we give an instance of CPLP where the inequality $Z_{D/TC} \leqslant Z_C$ is strict.

$$(c_{ij}) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad f = (3 \quad 3 \quad 2 \quad 2),$$

$$s = (2.4 \quad 2.4 \quad 1.6 \quad 1.6),$$

$$d = (1 \quad 1 \quad 1 \quad 1).$$

It is easy to check by enumeration that $Z = 6$. Next, we show that $Z_C = 6$ as well. In order to satisfy (T), fixed costs of at least 5 must be incurred. So, if $Z_C < 6$, the corresponding optimum solution satisfies $y_1 = y_3 = 1$ and $y_2 = y_4 = 0$ (or, by symmetry, one of the other three combinations yielding fixed costs of 5). But then, (D) and (B) imply that $x_{11} + x_{13} = 1$ and therefore transportation costs contribute an additional unit to total costs, showing that $Z_C = 6$. We will show that $Z_{D/TC} < Z_C$ for this instance by showing $Z_{D/TC} \leqslant 5.4$. This follows from the observation that the point $(\bar{x}, \bar{y})$ below belongs to Conv(CINTB) $\cap$ Conv(DINB).

$$(\bar{x}_{ij}) = \begin{pmatrix} 0.1 & 0.5 & 0 & 0.4 \\ 0.1 & 0.5 & 0.4 & 0 \\ 0.5 & 0.1 & 0 & 0.4 \\ 0.5 & 0.1 & 0.4 & 0 \end{pmatrix},$$

$$\bar{y} = (0.5 \quad 0.5 \quad 0.5 \quad 0.5).$$

To see that this point belongs to Conv(CINTB), note that it is the convex combination of the following four points (each with a multiplier of 0.25).

$$(x_{ij}^1) = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$y^1 = (0 \quad 1 \quad 0 \quad 1),$$

$$(x_{ij}^2) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0.4 & 0.6 & 0 \end{pmatrix},$$

$$y^2 = (0 \quad 1 \quad 1 \quad 0),$$

$$(x_{ij}^3) = \begin{pmatrix} 0.4 & 0 & 0 & 0.6 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$y^3 = (1 \quad 0 \quad 0 \quad 1),$$

$$(x_{ij}^4) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix},$$

$$y^4 = (1 \quad 0 \quad 1 \quad 0).$$

To see that the point $(\bar{x}, \bar{y})$ belongs to Conv(DINB), note that it is the convex combination of the following four points, where the first two have a multiplier of 0.1 and the last two have a multiplier of 0.4 in the convex combination.

$$(x_{ij}^1) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$y^1 = (1 \quad 0 \quad 1 \quad 0),$$

$$(x_{ij}^2) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$y^2 = (0 \quad 1 \quad 0 \quad 1),$$

$$(x_{ij}^3) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$y^3 = (0 \quad 0 \quad 1 \quad 1),$$

$$(x_{ij}^4) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$y^4 = (1 \quad 1 \quad 0 \quad 0).$$

This shows that

$$(\bar{x}, \bar{y}) \in \text{Conv(CINTB)} \cap \text{Conv(DINB)}.$$

Since $c\bar{x} + f\bar{y} = 5.4$, the proof of (2.24) is complete.

Finally consider (2.25). $Z_{D/BC}$ and $Z_{D/TBC}$ are at least as strong as Z_D. To show that they are equal to it, it suffices to show that the feasible set for $Z_{D/BC}$, namely Conv(DINT) $\cap$ Conv(BCINT), is identical to the feasible set for Z_D, namely P(D) $\cap$ Conv(BCINT). This follows from

$$\mathrm{Conv(DINT)} = P(\mathrm{D}) \cap \mathrm{Conv(INT)}$$

and

$$\mathrm{Conv(BCINT)} \subseteq \mathrm{Conv(INT)}.$$

Now consider $Z_{TD/BC}$. It has feasible set Conv(TDIN) $\cap$ Conv(BCIN). The equalities

$$\mathrm{Conv(TDIN)} = P(\mathrm{D}) \cap \mathrm{Conv(INT)}$$

and

$$\mathrm{Conv(BCIN)} = P(\mathrm{BC}) \cap \mathrm{Conv(IN)}$$

together with (2.9) imply $Z_{TC/BC} = Z_D$. □

Theorems 1 and 2 identify eight potentially different lower bounds for Z. The next obvious question to ask is about the expected tightness of the various bounds. In Section 4, we will provide our computational experience on a set of test problems. Another approach that has been gaining ground in recent years is probabilistic analysis. It requires probabilistic assumptions on the data, here c_{ij}, f_j, d_i, and s_j for all i, j. Then, the various bounds become random variables and they can be compared to the optimum Z, also a random variable, as the number of clients becomes large. We performed such an analysis in an earlier version of this paper and found that, under some probabilistic assumptions, the bounds fall into three groups. The bounds Z_C^T, $Z_{D/TC}$ and Z_C become arbitrarily close to Z as the number of clients increases. All three bounds Z^I, Z_D', and Z_D tend to 0.998 Z. The bounds Z^{BI} and Z_C^B are both very weak. Some of the important features of our probabilistic assumptions were that the transportation cost c_{ij} is proportional to the Euclidean distance between points i and j (the points are distributed uniformly in some region of the plane); the plant capacities and fixed costs are independent of j, i.e., $s_j = s$ and $f_j = f$; in addition, the client demands and plant capacities have the property that each d_i is small relative to s and s is small relative to $\sum_i d_i$. Details can be found in [6].

3. Computational complexity

In Section 2, eight bounds were obtained for CPLP, namely Z^{BI}, Z^I, Z_D', Z_D, Z_C^B, Z_C^T, $Z_{D/TC}$ and Z_C. Here we consider the computational complexity of calculating these bounds. We also consider the complexity of computing the optimum value Z of CPLP itself.

Theorem 3.

Z^{BI}, Z^I and Z_D' can be computed in polynomial time. (3.1)

Z_D and Z_C^B can be computed in pseudo-polynomial time. (3.2)

Computing Z_C^T, $Z_{D/TC}$, Z_C or Z is strongly NP-hard. (3.3)

Proof. Linear programs can be solved in polynomial time, so (3.1) holds for Z^{BI} and Z^I. Z_D' is obtained by minimizing $cx + fy$ over

$$P(\mathrm{D}) \cap \mathrm{Conv(CBINT')} = P(\mathrm{DCB}) \cap \mathrm{Conv(INT')}.$$

(The equality holds by the same argument as for (2.8).) The constraint set (N), (T′) is totally unimodular. So computing Z_D' reduces to the solution of a linear program. This proves (3.1).

Computing Z_D or Z_C^B is NP-hard since, when $c = 0$, the problem is nothing but a knapsack problem. Next we show that Z_D can be computed in pseudo-polynomial time. To get Z_D, we optimize $cx + fy$ over P(DCB) $\cap$ Conv(INT). It follows from a result of Grötschel, Lovász and Schrijver [16] that optimizing over P(DCB) $\cap$ Conv(INT) requires at most pseudo-polynomial time since optimizing over P(DCB) is polynomial and optimizing over Conv(INT) is pseudo-polynomial. To get Z_C^B, we optimize $cx + fy$ over P(DC) $\cap$ Conv(INT). So the same argument can be used. This proves (3.2).

Finally, (3.3) follows from the fact that the simple plant location problem is strongly NP-hard and polynomially reduces to the CPLP by setting the capacities s_j large enough. □

Although the CPLP is NP-hard, there are some special cases that can be solved in polynomial

time. In the next theorem, we show that the CPLP can be solved in polynomial time when the capacities and the demands are taken from the set $\{1, 2\}$ and it is NP-hard when the capacities are taken from the set $\{1, \ldots, p\}$, $p \geqslant 3$ even if all the demands are equal to 1. We also consider the complexity of computing Z_{D}, $Z_{\mathrm{C}}^{\mathrm{B}}$, $Z_{\mathrm{C}}^{\mathrm{T}}$, $Z_{\mathrm{D/TC}}$ and Z_{C} under the last assumption.

Theorem 4. (A) *The CPLP is strongly NP-hard when the capacities are taken from the set* $\{1, \ldots, p\}$ *for any fixed* $p \geqslant 3$, *even if all demands are equal to* 1. *Under this assumption,* Z_{D} *or* $Z_{\mathrm{C}}^{\mathrm{B}}$ *can be computed in polynomial time but computing* $Z_{\mathrm{C}}^{\mathrm{T}}$, $Z_{\mathrm{D/TC}}$ *or* Z_{C} *is strongly NP-hard.*

(B) *The CPLP can be solved in polynomial time when the capacities and demands are taken from the set* $\{1, 2\}$.

Proof. We will prove that the CPLP is strongly NP-hard for $p \geqslant 3$ by reducing the 3-dimensional matching problem (3DM) to CPLP.

The 3DM is defined by three distinct sets I, J, K, each of cardinality r, and a family F of triplets (i, j, k) where $i \in I$, $j \in J$ and $k \in K$. The question is whether there exists $S \subseteq F$ of cardinality r such that the elements of I, J, K each appear exactly once in the triplets of S. This problem is NP-complete [13]. We construct a CPLP as follows. There are $3r$ clients corresponding to the elements of I, J, K and r^3 facilities corresponding to all the triplets (i, j, k) with $i \in I$, $j \in J$, $k \in K$. The transportation costs from a triplet $(i, j, k) \in F$ to each of the clients i, j and k are equal to 0. All other transportation costs are equal to 1. The capacities are equal to 3, the demands to 1 and the fixed costs to $\varepsilon < 1/r$. The 3DM has a solution if and only if the above CPLP has a solution with value $r\varepsilon$.

To see that Z_{D} or $Z_{\mathrm{C}}^{\mathrm{B}}$ can be computed in polynomial time, note that $0 \leqslant \sum_i d_i \leqslant \sum_j s_j \leqslant pn$, where n denotes the number of facilities. Since p is fixed, a dynamic programming algorithm for solving the knapsack problem defined by (T) is polynomial. Now the result follows from [16].

To see that $Z_{\mathrm{C}}^{\mathrm{T}}$, $Z_{\mathrm{D/TC}}$ and Z_{C} are NP-hard, note that the transformation of the 3DM described above is valid even if we relax the capacity constraints (C) and (T) completely. The 3DM has a solution if and only if Z^{TC} has the value $r\varepsilon$. This completes the proof of part (A).

Now we prove part (B). From an instance of the CPLP where the demands and capacities are taken from $\{1, 2\}$, we construct a graph as follows. For each customer i with demand d_i we construct d_i nodes, for each facility j with capacity s_j we construct s_j nodes and for each facility j with capacity 1 we construct an additional node. The corresponding node sets are denoted by U, V and W respectively. The graph has an edge with cost c_{ij} joining each node of U associated with customer i to each node of V associated with facility j. For each facility j such that $s_j = 2$, the two nodes of V associated with j are joined by an edge with cost $-f_j$. Finally, each node of V associated with a facility with $s_j = 1$ is joined to the corresponding node of W by an edge of cost $-f_j$. Now the CPLP is equivalent to solving a minimum cost matching problem in this graph, where each node of U is required to be exactly matched and each node of $V \cup W$ must meet at most one edge of the matching. (The closed facilities correspond to the nodes of V that are matched to a node of W or to another node of V.) Part (B) follows from the fact that the matching problem can be solved in polynomial time [9]. □

It is interesting to note that, although Z can be computed in polynomial time under the assumption made in Theorem 4, part (B), it is not clear whether Z_{C} and $Z_{\mathrm{C}}^{\mathrm{T}}$ can. In fact, there are values of the Lagrange multipliers for which the Lagrangian relaxations $Z_{\mathrm{C}v}$ and $Z_{\mathrm{C}v}^{\mathrm{T}}$ are NP-hard to solve. (For example, the reduction of the 3DM used for proving Theorem 4, part (A) is still valid when $v = 0$.)

4. Computational results

Our computational study compares the relaxations Z^{I}, Z_{D}, $Z_{\mathrm{C}}^{\mathrm{T}}$ and Z_{C} to the optimum value Z of CPLP. We also compare a Lagrangian heuristic to greedy and interchange heuristics. One objective of the study is to test the tightness of the bounds as a function of parameters of the problem. In particular, we consider the influence of tight capacity constraints.

Z_{D} is computed using a subgradient algorithm stopped after a maximum of 200 iterations. The knapsack subproblems are solved using the algorithm of Fayard and Plateau [11]. To compute

Z^{I}, we solve the Lagrangian dual giving $Z^{\mathrm{T}}_{\mathrm{DC}}$ by a subgradient algorithm (recall that $Z^{\mathrm{I}} = Z^{\mathrm{T}}_{\mathrm{DC}}$). To obtain $Z^{\mathrm{T}}_{\mathrm{C}}$, Z_{C} and Z, we implement the cross decomposition of Van Roy [34]. The optimum value Z is computed using a branch-and-bound algorithm based on the bound Z_{C}. The bound Z_{C} itself is calculated by solving iteratively a sequence of problems $Z_{\mathrm{C}v}$. For each value of v, $Z_{\mathrm{C}v}$ is obtained using a branch-and-bound algorithm based on $Z^{\mathrm{T}}_{\mathrm{C}v}$. Finally, the simple plant location problem that yields $Z^{\mathrm{T}}_{\mathrm{C}v}$ is solved using the dual ascent procedure of Erlenkotter [10] which also involves branch and bound. Clearly, due to the triply nested branch-and-bound algorithms, cross decomposition is most effective when the duality gap $Z - Z^{\mathrm{T}}_{\mathrm{C}}$ is small. Dual heuristics are needed to accelerate the search for the optimal Lagrange multipliers. Lacking this, it is not obvious how to generate good dual information from the primal subproblems. For the harder problem instances, the cross decomposition method often reduces to a rather inefficient column generation method, as corroborated by Van Roy.

On the heuristic side, we implemented DROP-HI and VSM, a greedy heuristic starting from all facilities open and an interchange heuristic, respectively, as described in [21]. The implementation suggested by Khumawala [22] gives similar,

Table 1
$\Sigma_j s_j = 1.5\ \Sigma_i d_i$

Relative error (%)	Relaxations				Heuristics		
	Z_{C}	Z_{D}	$Z^{\mathrm{T}}_{\mathrm{C}}$	Z^{I}	H	VSM	DROP – HI
25 × 8	0	0	5.70	5.88	0	0	17.27
	0	0.01	6.38	6.85	0	5.73	5.73
	0	0.02	5.85	6.67	0	6.22	11.14
	0	0	1.30	5.10	0	13.13	13.13
	0	0.02	5.48	5.71	0	8.82	8.82
25 × 16	0.37	0.37	3.05	3.39	0	15.69	15.69
	0	0	3.12	14.32	0	6.23	20.80
	0	0	2.14	15.07	0	13.30	13.30
	0.12	0.12	1.63	2.49	0	10.92	15.20
	0.01	0.07	3.09	20.87	0	0.29	5.36
25 × 25	0	0.01	1.41	46.88	0	5.95	30.01
	0	0.01	0.54	2.05	0	10.02	20.96
	0.04	0.09	1.65	2.30	0	9.18	26.96
	0	0	1.93	15.28	0	15.01	38.25
	0.07	0.09	1.55	29.27	0	12.57	26.36
50 × 16	0	0.01	2.18	4.30	0	15.30	17.97
	0	0	0.42	1.23	0	17.75	23.71
	0	0.01	2.27	2.45	0	4.08	24.42
	0.26	0.29	2.07	2.10	0	2.77	16.78
	0	0	0.91	9.27	0	10.48	22.86
50 × 33	0.10	0.14	1.48	3.49	0	25.52	26.24
	0.05	0.10	1.17	17.33	0	19.35	24.10
	0	0.16	0.18	0.59	0	18.58	28.02
	0.12	0.14	1.03	3.01	0	5.77	26.10
	0.05	0.09	1.61	2.89	0	11.72	29.98
50 × 50	0	0.06	0.25	1.45	0	[a]	27.63
	0	0.07	0.09	19.62	0	[a]	26.08
	0	0.02	0.38	32.96	0	[a]	24.44
	0.01	0.02	0.19	0.41	0	[a]	28.88
	0	0.10	0	11.80	0	[a]	27.58
Average (%)	0.04	0.07	1.97	9.83	0	10.57	21.46

[a] Time limit of 120 seconds on DEC 20 exceeded.

although slightly inferior, results. We also consider a Lagrangian heuristic, denoted by (H). At each step of the subgradient algorithm for computing Z_D, a solution (x_D, y_D) of the Lagrangian subproblem is found. If it is feasible for the original problem, it is an optimum solution. Otherwise, consider y_D as the set of open facilities and solve a transportation problem to find a feasible solution of the CPLP, say $(\bar{x}_D, y_D)$. The best such solution found during the subgradient iterations is taken as the solution of heuristic (H). We have observed that, typically, only few different vectors y_D are generated during the algorithm and so only few transportation problems have to be solved. We used the code of Srinivasan and Thompson [32] for solving the transportation problems.

To validate our algorithms, we solved the classical test problems of Kuehn and Hamburger [25] and were able to confirm the computational results already reported in the literature. For example our evaluation of $(Z - Z_C)/Z$ coincides with ε^{OPT} in Table II of Van Roy [34] in all but one of the instances. (In problem 24, we find 0.18% instead of 0.21% but Van Roy truncated his branch-and-bound after 2000 nodes whereas we completed the enumeration.) Van Roy does not report Z_C^T. We find that $Z_C^T = Z_C$ for all the problems except problems 11 and 12, where $(Z -$

Table 2
$\Sigma_j s_j = 2\,\Sigma_i d_i$

Relative error (%)	Relaxations				Heuristics		
	Z_C	Z_D	Z_C^T	Z^I	H	VSM	DROP−HI
25× 8	0	0	5.94	6.22	0	0	24.05
	0.75	0.79	10.31	10.32	0	0.82	8.40
	0	0.05	7.48	7.80	0	4.27	14.40
	0	0.02	2.23	2.25	0	0	35.71
	0	0	5.24	5.26	0	19.54	21.84
25×16	0.19	0.19	4.91	4.97	0	10.36	41.98
	0.04	0.09	1.43	1.47	0	13.00	22.36
	0	0	2.09	2.27	0	8.75	38.64
	0.15	0.15	4.36	4.36	0	14.30	20.91
	0	0	2.74	3.34	0	18.59	44.04
25×25	0.05	0.12	2.28	2.28	0	4.82	34.68
	0.18	0.23	1.42	1.57	0	9.56	29.89
	0	0.04	0.78	0.84	0	20.25	40.33
	0	0.03	1.76	2.38	0	13.30	37.25
	0.11	0.12	3.04	3.07	0	6.76	41.36
50×16	0	0.19	3.66	3.89	0	13.19	27.92
	0.01	0.17	2.55	2.57	0	5.93	23.11
	0	0	0.46	0.49	0	11.05	25.08
	0.42	0.43	2.45	11.85	0	14.12	14.12
	0	0	0	0	0	7.61	41.27
50×33	0.04	0.04	1.50	1.50	0	10.84	46.67
	0.09	0.09	1.15	1.19	0	11.81	32.34
	0.08	0.10	1.52	1.86	0	19.98	47.39
	0.01	0.02	0.67	0.81	0	14.38	28.48
	0.09	0.09	1.34	1.34	0	12.82	39.08
50×50	0	0	0.23	0.34	0	[a]	36.27
	0.21	0.27	1.01	1.25	0.01	[a]	41.60
	0	0.02	0.14	1.43	0	[a]	45.38
	0	0.03	0.13	0.36	0	[a]	39.02
	0	0.04	0.02	0.13	0	[a]	42.78
Average (%)	0.08	0.11	2.43	2.91	0	10.64	33.21

[a] Time limit of 120 seconds on DEC 20 exceeded.

$Z_C^T)/Z = 0.074\%$ and 0.277% respectively. Regarding Z_D, our results are better than expected. The reason is that, although Nauss [27] recommends to compute the bound Z_D in his paper, he reports computational results that are not based on Z_D. More specifically, Nauss relaxes the integrality requirements in the knapsack subproblems needed to compute Z_D (see first sentence of second paragraph in section "Computational results" of [27]). The reason for doing this is unclear, since the knapsack subproblems are very easy to solve when all the capacities s_j are equal. As a result, Nauss computes Z^I instead of Z_D. Indeed, in Table I of [27], the gaps closed by LGR1 and LGR2 after subgradient optimization are very similar. (By Theorem 1, they should eventually be identical, after sufficiently many steps of the subgradient algorithm.) By solving true knapsack subproblems, we find that Z_D often greatly improves on Z^I. For example, $Z_D = Z$ for problems 6, 8 and 9 as numbered by Nauss (Nauss reports gaps for these problems). Going back to the numbering of Van Roy, we find that $Z_D = Z$ for problems 1–6. For the more tightly capacitated problems 7–12, we find $(Z - Z_D)/Z = 0.06\%$, 0.20%, 0.64%, 0.95%, 0.66% and 0.36% respectively. For these 12

Table 3
$\Sigma_j s_j = 3\,\Sigma_i d_i$

Relative error (%)	Relaxations				Heuristics		
	Z_C	Z_D	Z_C^T	Z^I	H	VSM	DROP-HI
25× 8	0	0	7.06	8.94	0	0	12.85
	0	0	1.47	1.71	0	12.10	22.81
	0	0	5.69	5.91	0	0	3.92
	0	0	4.98	5.07	0	8.97	16.11
	0	0	4.13	4.63	0	10.38	30.23
25×16	0	0.04	6.00	6.10	0	2.90	39.29
	0	0	1.45	1.47	0	9.65	39.18
	0	1.48	4.38	8.44	0	3.13	39.35
	0	0.25	5.44	5.54	0	0	60.09
	0	0.09	3.65	3.65	0	0	39.47
25×25	0.11	0.11	0.39	0.41	0	16.97	62.02
	0.20	0.33	4.60	4.63	0.09	0.69	53.54
	0	0.06	0.45	0.45	0	1.56	53.93
	0.92	1.19	2.06	2.09	0	17.82	52.53
	0	0.44	2.18	2.26	0	0	61.60
50×16	0.19	0.82	4.10	4.71	0	6.90	33.88
	0.13	1.55	5.55	5.66	0	0.63	15.29
	0	0	1.34	1.41	0	0	26.03
	0	0	1.95	2.40	0	3.35	26.83
	0	0	2.08	2.08	0	0	41.84
50×33	0.03	0.46	2.40	2.51	0	11.40	53.72
	0.56	0.70	2.90	2.93	0	12.37	54.61
	0.76	0.92	1.53	1.59	0	8.38	47.45
	0.27	0.30	0.76	0.79	0	13.02	40.10
	0.41	0.75	2.06	2.06	0	2.35	53.66
50×50	0.17	0.26	1.43	1.43	0	4.20	64.06
	0.52	0.55	1.14	1.16	0	[a]	62.71
	0.29	0.51	0.88	0.88	0	11.71	44.40
	0.30	0.35	0.52	0.56	0	[a]	49.08
	0.24	0.39	0.71	0.73	0	[a]	44.47
Average (%)	0.16	0.38	2.78	3.07	0	5.87	41.50

[a] Time limit of 120 seconds on DEC 20 exceeded.

instances, the heuristic (H) always finds the optimum solution. The heuristics DROP-HI and VSM perform reasonably well (see [21]).

To complement these results, we ran the algorithms on a different and larger set of test problems.

We generated 5 sets of 30 problems. In each set, there are 5 problems of each of the following sizes: 25×8, 25×16, 25×25, 50×16, 50×33, 50×50 where the first number is the number of clients and the second is the number of facilities. The transportation costs are computed by generating points representing the clients and facilities uniformly in a unit square. The Euclidean distance between them is multiplied by 10 to define unit transportation costs. The demands are generated from $U[5,35]$, where $U[a, b]$ denotes the uniform distribution in the interval $[a, b)$. The capacities s_j are generated from $U[10, 160]$ and the fixed costs by the formula

$$f_j = U[0, 90] + U[100, 110]\sqrt{s_j}$$

to reflect economies of scale. From the above data, 5 different sets of problems are generated by scaling the capacities using the ratios $\Sigma_j s_j / \Sigma_i d_i =$ 1.5, 2, 3, 5, 10. Finally f_i is doubled for the first 2 sets. Although the above choices are somewhat arbitrary, our objective was to generate a reasonably difficult set of test problems. The actual data are available from the authors.

Table 4
$\Sigma_j s_j = 5\,\Sigma_i d_i$

Relative error (%)	Relaxations				Heuristics		
	Z_C	Z_D	Z_C^T	Z^I	H	VSM	DROP-HI
25×8	0	0	11.61	11.75	0	0	3.92
	0	0	5.95	5.95	0	3.98	10.64
	0	0.31	5.39	5.65	0	0	18.88
	0	2.28	11.40	11.66	0	0	11.90
	0	0	0	0	0	0	23.92
25×16	0	0	1.57	2.07	0	0	59.50
	0	0	0.22	0.82	0	1.61	49.81
	0	0	3.72	4.06	0	4.55	37.36
	0	0	1.76	2.01	0	3.18	25.06
	0.12	0.12	1.54	2.34	0	0	40.53
25×25	0	1.07	3.76	3.89	0	3.33	52.94
	0.19	0.74	2.68	12.10	0	13.31	32.80
	0.27	2.57	5.90	5.90	0.07	7.11	59.22
	0.30	0.53	2.43	2.90	0	0	64.85
	0	0.22	2.11	2.20	0	2.22	34.07
50×16	0	0	0.51	1.40	0	0	30.36
	0	0.70	3.27	3.68	0	2.37	28.91
	0.38	0.38	1.87	3.92	0	0	31.11
	0.65	2.10	4.73	4.90	0	0	22.01
	0	0.85	4.13	4.31	0	1.76	25.55
50×33	0	0.53	1.79	1.87	0	0	50.15
	0	0.01	0.81	0.91	0	5.46	68.73
	0	0.29	1.25	1.27	0	0	52.80
	0.07	0.15	0.92	1.17	0	1.93	69.54
	0.34	1.16	2.12	2.20	0	3.78	45.31
50×50	0.35	0.51	1.27	1.30	1.13	8.56	64.41
	0.41	0.34	2.80	2.84	1.24	0.92	79.34
	0.22	0.82	2.45	2.45	0.01	17.38	58.11
	0.01	0.43	0.89	0.89	0.42	0.26	67.37
	0.48	0.68	2.48	2.48	2.02	1.37	52.14
Average (%)	0.13	0.59	3.04	3.63	0.16	2.77	42.37

We report our computational experience with these 5 sets of problems in Tables 1–5. In each case we give the relative error of the bound to the optimum value. For a relaxation such as Z_C we report $(Z - Z_C)/Z$, and for a heuristic with value $Z(H)$ we report $(Z(H) - Z)/Z$.

Some remarks can be made about Tables 1–5. The relaxations Z_C and Z_D are both significantly stronger than Z_C^T and Z^I for this set of test problems. Z_C is about twice as strong as Z_D for the tightly capacitated problems and this advantage increases for the less capacitated ones. The quality of both bounds deteriorates in the intermediate range, i.e. $\Sigma_j s_j/\Sigma_i d_i = 3$ and 5, but Z_C never exceeds 1% and Z_D never exceeds 3%. Whereas for the Kuehn and Hamburger test problems Z_C^T is frequently equal to Z_C, here Z_C^T is typically much weaker. For the tightly capacitated problems, the negative effect of removing the knapsack constraint (T) is more pronounced when there are few facilities (e.g. 8 or 16). Our subgradient algorithm for computing Z^I had difficulties converging for the tightly capacitated problems of Table 1. So the large gaps appearing there should be interpreted with caution. DROP-HI and VSM are the two heuristics recommended by Jacobsen, based on tests with the Kuehn and Hamburger problems. Their performance is very different when the facilities do not have identical capacities and fixed costs, as pointed out by Domschke and

Table 5
$\Sigma_j s_j = 10\,\Sigma_i d_i$

Relative error (%)	Relaxations				Heuristics		
	Z_C	Z_D	Z_C^T	Z^I	H	VSM	DROP-HI
25× 8	0	0	6.06	6.10	0	0	17.68
	0	0	0	0.31	0	0	24.90
	0	0	0	0	0	0	22.09
	0	0	6.04	7.31	0	2.10	3.32
	0	0.21	3.51	3.69	0	0	1.91
25×16	0	0	6.20	6.35	0	0	22.37
	0	0	8.24	9.82	0	0	35.20
	0	0	9.08	9.39	0	0	51.02
	0	0	0	0.61	0	0	58.71
	0	0	0	0	0	0	34.97
25×25	0	0	1.12	1.21	0	0	48.54
	0	0.43	10.41	11.32	0	5.39	64.99
	0.05	0.08	3.74	3.74	0	3.05	41.00
	0	0	5.35	5.90	0	1.55	51.62
	0	0	5.36	8.78	0	8.41	80.28
50×16	0	0.13	4.13	4.64	0	0	18.02
	0	1.24	6.31	7.77	0	0	7.52
	0	2.02	4.46	6.26	0	0	19.63
	0	0	0	0	0	0	21.89
	0	0	1.80	2.37	0	0	27.11
50×33	0	0	2.23	14.10	0	10.50	62.96
	0	0	0.27	0.29	0	0	50.59
	0	0.13	2.33	5.36	0	0	37.67
	0	0.72	1.77	1.88	0	7.98	48.65
	0	0	0.35	0.49	0	0.94	58.59
50×50	0.01	0.51	2.31	10.94	0	1.07	77.06
	0	0	0.81	0.94	0	10.22	81.07
	0.04	0.83	3.50	4.32	0	0	76.16
	0.02	0.30	1.56	2.01	0.10	0	75.74
	0.05	0.90	3.85	4.59	0.22	5.62	63.11
Average (%)	0.01	0.25	3.36	4.68	0.01	1.88	42.81

Table 6
Computing times (average of 25 problems)

Problem Size	Seconds on IBM 3081		Second on DEC 20			
	Z_C	Z_C^T	Z^I	Z_D and Z_H	VSM	DROP-HI
25× 8	0.9	1.0	2.1	1.7	0.6	0.1
25×16	14	12	5.7	3.9	2.8	0.2
25×25	140	69	11	9.0	9.6	0.5
50×16	34	23	8.7	9.6	5.7	0.5
50×33	1900	570	28	23	41	1.7
50×50	6800	5200	62	37	> 92 [a]	3.7

[a] 13 instances exceeded the time limit of 120 seconds on a DEC 20.

Drexl [8]. Our own computational experience serves to reiterate this warning. DROP-HI tends to close the large facilities first, clearly not a good strategy. VSM performed reasonably well on the less capacitated problems. However, the clear winner is heuristic (H), failing to find the optimum solution in only 10 out of 150 cases. This is particularly remarkable considering that the time needed to compute Z_D and to find the heuristic solution (H) only grows moderately with problem size. By contrast, VSM exceeds the time limit of 120 seconds on a DEC 20 in thirteen instances. Computing times are provided in Table 6. More extensive computational results are reported in Sridharan [31].

6. Conclusion

In this paper we have studied relaxations of CPLP from three different angles: (i) inequalities among the corresponding bounds, (ii) computational experiments and (iii) complexity. The results in (i) can be thought of as a ready-reckoner to indicate the relaxation that is easiest to solve among those that yield the same bound. Eight genuinely different bounds were identified and ranked according to their tightness.

The weak LP relaxation can be solved analytically but the bound is too weak to be used in the solution of medium or large problems. The strong LP relaxation and the relaxation proposed by Nauss [27] are both relatively easy to solve and provide useful bounds. The relaxation proposed by Van Roy [34] is extremely tight, but this relaxation is strongly NP-hard to compute. Finally, the bound proposed by Guignard and Kim [19], although also strongly NP-hard to compute, warrants further investigation.

The computational results revealed that, for Z_C and Z_D, problems that are either loosely or tightly capacitated have smaller duality gaps than those in the intermediate range. Based on our computational results we suggest the use of a Lagrangian heuristic (H) to solve large instances of the CPLP. It provides considerably better solutions than the VSM heuristic and is also superior on computational time. Furthermore, it is obtained in conjunction with a lower bound.

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